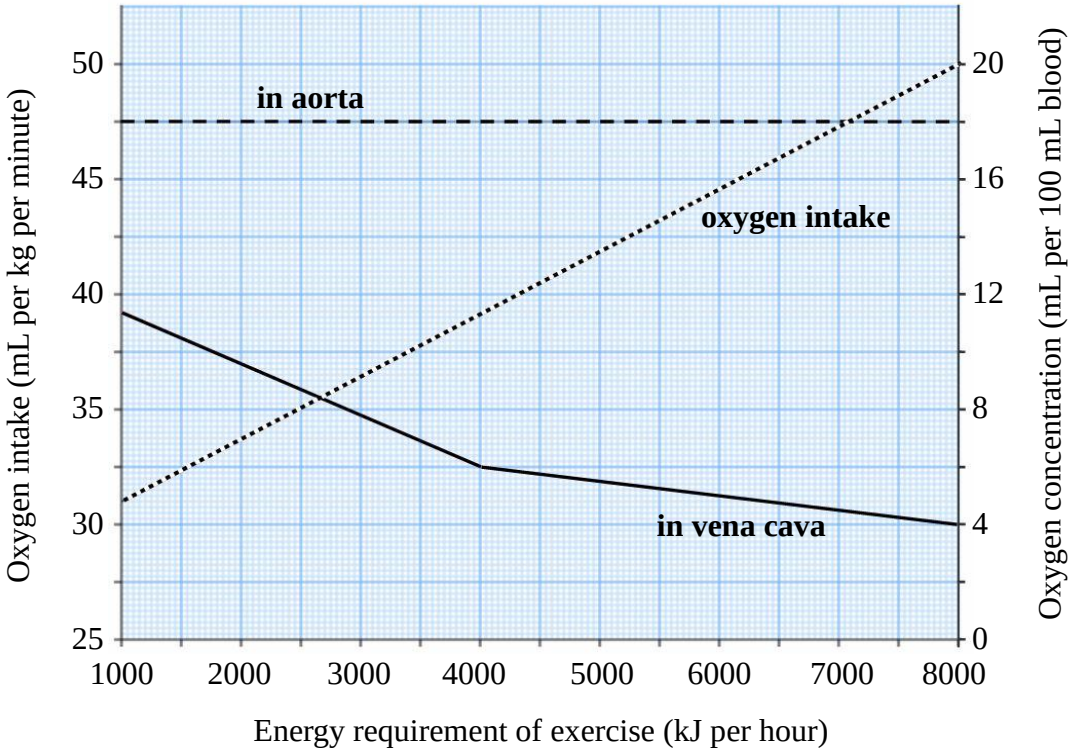
Structured questions

|!|BCELB0340000001|!| ★

The graph below shows the oxygen intake, the blood oxygen concentrations in the aorta and the vena cava of a person performing exercise with varying energy requirement.



(a) Energy requirement of running is 2800 kJ per hour. Determine the blood oxygen concentration in the aorta and the vena cava for running from the graph. (2 marks)

(b) Based on the graph, describe how the oxygen intake changes with exercise of varying energy requirement. (1 mark)

(c) Explain how the high blood oxygen concentration in the aorta can be maintained as exercise intensity increases. (3 marks)

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(d) Describe the physiological process that leads to the high blood oxygen concentration in the aorta. (4 marks)

##

(a) Blood oxygen concentration in the aorta is 18 mL per 100 mL blood. (1)
Blood oxygen concentration in the vena cava is 8.2 mL per 100 mL blood. (1)

(b) As the energy requirement of exercise increases, the oxygen intake increases. (1)

(c) During vigorous exercise, the rate and depth of breathing increase. (1)
More oxygen enters the air sacs per unit time. (1)
The diffusion gradient of oxygen between the air sacs and the blood becomes steeper / The rate of oxygen diffusion into the blood increases, (1)
thus the blood oxygen concentration of the aorta is maintained at a high level.

(d) During vigorous exercise, the respiration rate of skeletal muscles increases, releasing more energy and more carbon dioxide. (1)
Chemoreceptors in the carotid bodies, aortic bodies and respiratory centre detect the rise in the blood carbon dioxide content / the fall in blood pH and send nerve impulses to the respiratory centre in the medulla oblongata. (1)
The respiratory centre then sends more nerve impulses (via a sympathetic nerve) to the intercostal muscles and diaphragm muscle. (1)
This causes these muscles to contract more frequently and more forcefully. (1)
Therefore, the rate and depth of breathing increase.

(10 marks)

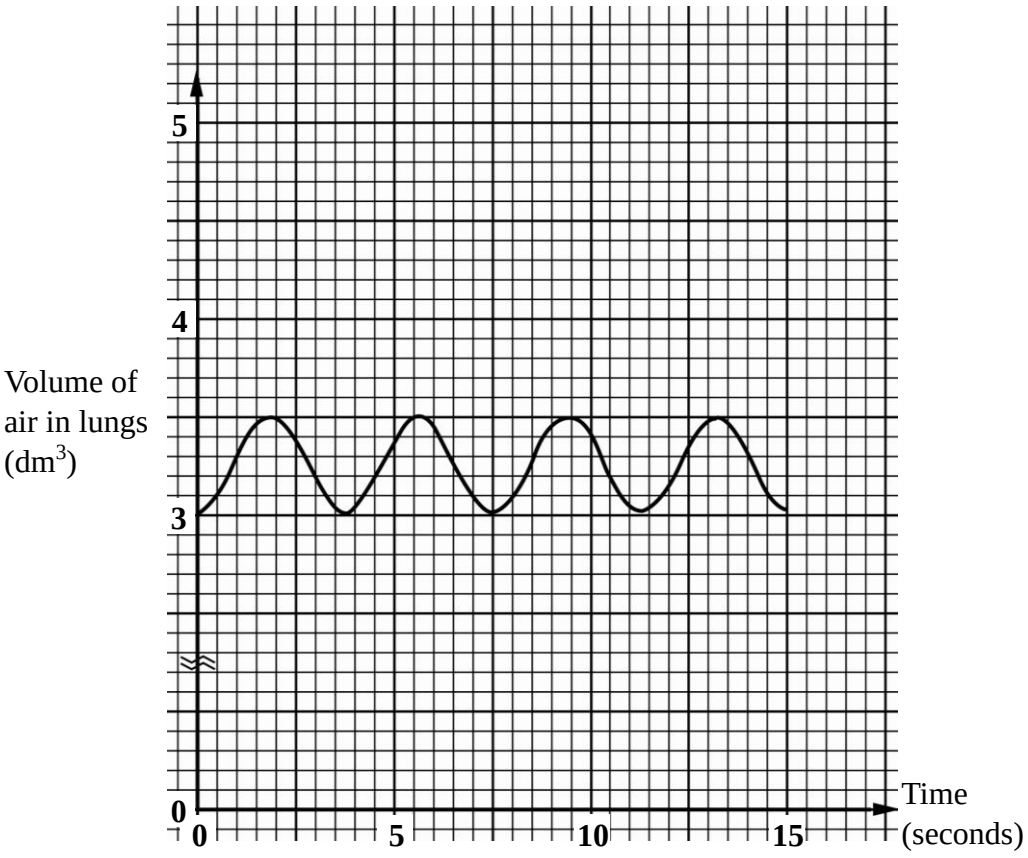
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##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Analysing data	1
(c)	Understanding basic concepts	3
(d)	Understanding basic concepts	4

|!|BCELB0340000002|!| ★

In an investigation, Aaron was asked to sit on a chair. The changes in the volume of air in his lungs were measured for 15 seconds. The graph below shows the results.



(a) Find out the volume of air breathed out by Aaron in one breath from the graph above. (1 mark)

_____

(b) Calculate Aaron's breathing rate at rest. Show your working. (2 marks)

_____

_____

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(c) Calculate Aaron's ventilation rate at rest. Show your working. (2 marks)

(d) (i) How would Aaron's ventilation rate change when he was doing vigorous exercise? (2 marks)

(ii) Describe how the sympathetic nerve brings about the change in ventilation rate during exercise. (2 marks)

##

(a) $0.5\ dm^3$ (1)

(b) Aaron's breathing rate

$= 4 \div 15 \times 60$ (1)

$= 16$ breaths per minute (1)

(c) Aaron's ventilation rate

$= 0.5\ dm^3 \times 16$ breaths per minute (1)

$= 8\ dm^3 / min$ (1)

(d) (i) During exercise, the rate and depth of breathing would increase. (1)

As a result, the ventilation rate would increase. (1)

(ii) The respiratory centre sends more nerve impulses via a sympathetic nerve to the intercostal muscles and diaphragm muscle. (1)

This causes these muscles to contract more frequently and more forcefully. (1)

Therefore, the rate and depth of breathing increase.

(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Mathematical skills	2
(c)	Mathematical skills	2
(d)(i)	Applying concepts	2
(d)(ii)	Applying concepts	2

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|!|BCELB0340000003|!| ★

In an investigation, Charles was asked to breathe in different types of gas mixtures (A, B and C), and then the researchers measured and recorded his breathing rate in the table below.

Gas mixture	**Composition**	**Breathing rate (number of breaths / min.)**
A	20% oxygen + 0.03% carbon dioxide	11
B	23% oxygen + 0.03% carbon dioxide	10
C	23% oxygen + 0.3% carbon dioxide	25

(a) With reference to data in the table, deduce oxygen or carbon dioxide has a greater effect on the breathing rate. (3 marks)

(b) Describe how a change in the content of the gas you have given in (a) would lead to an increase in the breathing rate. (4 marks)

(c) By breathing in which gas mixtures (A, B or C) can help a patient with breathing difficulty? Explain your choice. (3 marks)

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##

(a) When the carbon dioxide content increased from 0.03% to 0.3%, with the oxygen content remaining constant, the breathing rate increased by 150%. (1)

When the oxygen content decreased from 23% to 20%, with the carbon dioxide content remaining constant, the breathing rate increased by 10%. (1)

This shows that a small change in the carbon dioxide content led to a greater change in the breathing rate. (1)

(b) When the carbon dioxide content in blood increases, the blood pH decreases. (1)

The chemoreceptors in the carotid bodies, aortic bodies and respiratory centre are stimulated by the fall in blood pH. They send nerve impulses to the respiratory centre in the medulla oblongata. (1)

The respiratory centre then sends more nerve impulses (via a sympathetic nerve) to the intercostal muscles and diaphragm muscle. (1)

This causes the intercostal muscles and diaphragm muscle to contract more frequently and more forcefully. (1)

(c) Gas mixture C (1)

It has a higher carbon dioxide content which would lead to a fall in blood pH and in turn stimulates the patient to breathe faster and deeper. (1)

It also supplies more oxygen to the patient. (1)

(10 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
	Applying concepts	1
(b)	Applying concepts	4
(c)	Applying concepts	3

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|!|BCELC0340000004|!|

A student's rate and depth of breathing were measured at rest and just after exercise. The table below shows the results.

	Depth of breathing (cm^3)	**Rate of breathing (breaths per minute)**
At rest	400	20
Just after exercise	900	36

(a) Calculate the total volume of air breathed in per minute when the student was

(i) at rest. (1 mark)

(ii) just after exercise. (1 mark)

(b) Assuming that inhaled air contains 20% oxygen and exhaled air contains 16% oxygen, calculate the volume of oxygen entering the blood at the air sacs per minute when the student was

(i) at rest. (1 mark)

(ii) just after exercise. (1 mark)

(c) Based on the table, state the effect of exercise on the rate and depth of breathing. (2 marks)

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(d) In order to bring about the changes in (c), state

(i) the part of the brain involved; (1 mark)

(ii) the corresponding stimulus; and (1 mark)

(iii) ***two*** effectors responsible for the changes. (2 marks)

##

(a) (i) At rest: 400 × 20 = 8000 cm^3 per minute (1)

(ii) Just after exercise: 900 × 36 = 32400 cm^3 per minute (1)

(b) (i) At rest: 8000 × (20 – 16)% = 320 cm^3 per minute (1)

(ii) Just after exercise: 32400 × (20 – 16)% = 1296 cm^3 per minute (1)

(c) During exercise, the rate of breathing increases (1)

and the depth of breathing also increases. (1)

(d) (i) The medulla oblongata (1)

(ii) The blood pH falls / carbon dioxide content of blood increases (1)

(iii) The diaphragm muscles (1)

The intercostal muscles (1)

(10 marks)

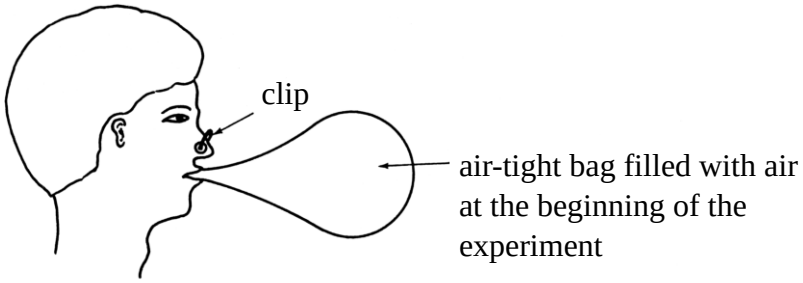
##

	Skills assessed	Marks
(a)(i)	Mathematical skills	1
(a)(ii)	Mathematical skills	1
(b)(i)	Mathematical skills	1
(b)(ii)	Mathematical skills	1
(c)	Analysing data	2
(d)(i)	Understanding basic concepts	1
(d)(ii)	Understanding basic concepts	1
(d)(iii)	Understanding basic concepts	2

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|!|BCELB0340000005|!| ★

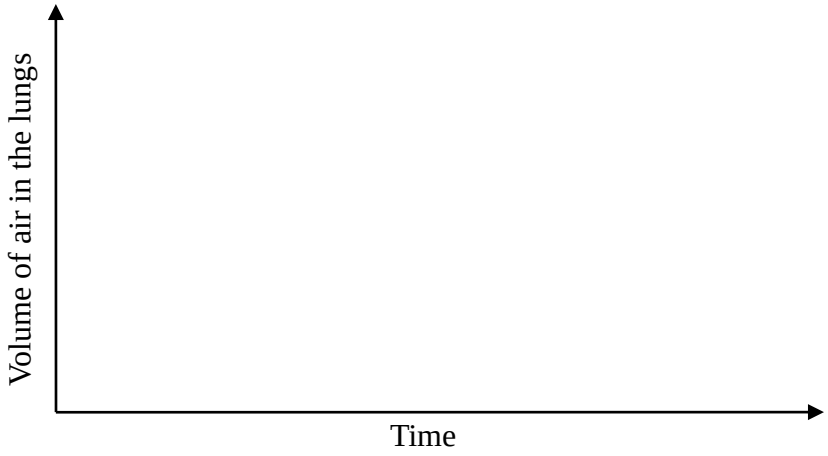
In an investigation, Tommy breathed into and out of an air-tight bag, with his nose clipped, as shown in the diagram below. The bag was filled with air at the beginning of the experiment.



(a) How would Tommy's rate and depth of breathing change during the experiment? (2 marks)

(b) Explain how the changes in (a) occurred. (6 marks)

(c) After 40 seconds, the clip and plastic bag were removed. Sketch in the space below to show how the volume of air in Tommy's lungs changes in the first 15 seconds after the removal of the clip and the plastic bag. (3 marks)

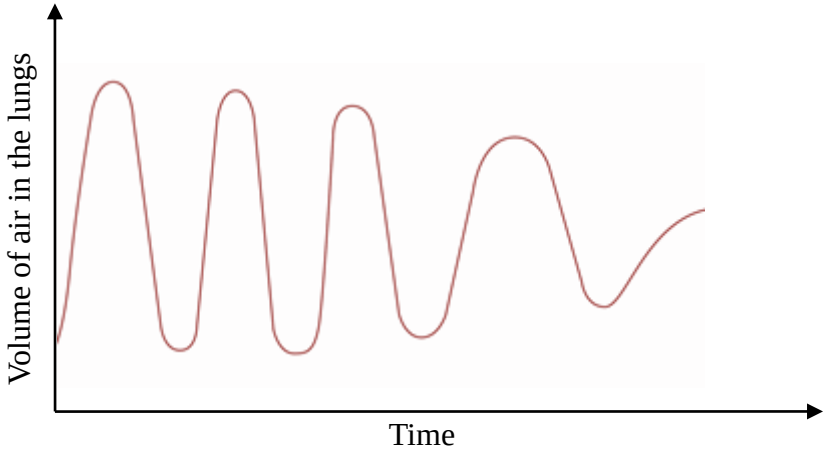


##

(a) His rate of breathing increased (1)
and his depth of breathing also increased. (1)

(b) As Tommy breathed into and out of the bag repeatly, the carbon dioxide content of the air in the bag increased. The inhaled air contained more carbon dioxide. (1)
The difference in carbon dioxide content between the blood and the inhaled air became smaller, the rate of carbon dioxide removal from the blood decreased, so more carbon dioxide accumulated in the blood. (1)
The blood pH fell and this stimulated the chemoreceptors in the carotid bodies and aortic bodies. (1)
The chemoreceptors then sent nerve impulses to the respiratory centre in the medulla oblongata. (1)
The respiratory centre then sent more frequent nerve impulses (via a sympathetic nerve) to the diaphragm muscle and intercostal muscles, (1)
causing them to contract more rapidly and strongly. (1)

(c)



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Curve rising up and down for one breath (1)

The rate of breathing decreases with time. (1)

The depth of breathing decreases with time. (1)

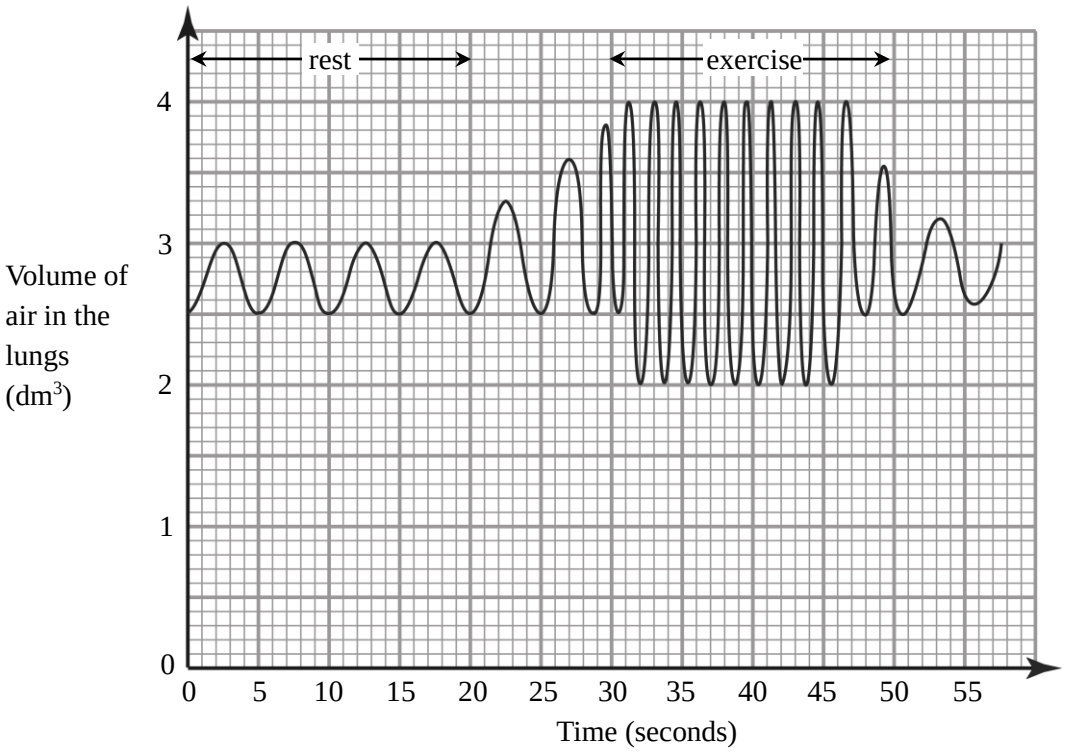
(11 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Applying concepts	6
(c)	Applying concepts	3

|!|BCELC0340000006|!|

The graph below shows the changes in the volume of air in the lungs of a person at rest and during exercise.



(a) What is the tidal volume when the person is at rest? (1 mark)

(b) Compare the tidal volume and rate of breathing when the person is at rest and during exercise. (2 marks)

(c) How are the changes in tidal volume and breathing rate brought about during exercise? (6 marks)

##

(a) 0.5 dm^3 (1)

(b) The tidal volume increased. (1)

The rate of breathing increased. (1)

(c) When a person is exercising, the rate of cellular respiration increases and carbon dioxide is produced at a faster rate. (1)

The carbon dioxide content of blood increases and blood pH decreases. (1)

The fall in blood pH stimulates chemoreceptors in the carotid and aortic bodies. (1)

The chemoreceptors then send nerve impulses to the respiratory centre in the medulla oblongata. (1)

The respiratory centre sends more nerve impulses (via a sympathetic nerve) to the intercostal muscles and diaphragm muscle, (1)

causing them to contract more frequently and more forcefully. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	2
(c)	Applying concepts	6

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|!|BCELB0340000007|!| ★ **Cross topic**

The diagram below shows an athlete running on a treadmill.



The table below shows the athlete's ventilation rate when he was at rest and running on a treadmill at different speeds.

Running speed (km / hr)	**Ventilation rate (dm^3 / min)**
0 (rest)	7.5
2	11
4	19
6	25
8	37
10	60

Ventilation rate measures the volume of air breathed in or out per minute.

(a) Ventilation rate can be calculated by multiplying two variables. What are the two variables? (2 marks)

(b) How does the ventilation rate change with running speed? What is the significance of this change? (4 marks)

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(c) (i) How do skeletal muscle cells cope with the energy demand when the oxygen supply cannot meet the demand for aerobic respiration? (1 mark)

(ii) Write a word equation to represent the overall process you have given in (c)(i). (1 mark)

(iii) Why does the ventilation rate remain high for a short period after exercise? (2 marks)

##

(a) Tidal volume (1)
Breathing rate (1)

(b) The ventilation rate increases as the running speed increases. (1)
When the running speed increases, skeletal muscles require more energy for increased muscular activities. (1)
An increased ventilation rate allows the body to obtain more oxygen for aerobic respiration to release more. (1)
It also allows faster removal of carbon dioxide produced from respiration. (1)

(c) (i) The skeletal muscle cells undergo anaerobic respiration to release extra energy to cope with the immediate energy need. (1)
(ii) glucose → energy + lactic acid (1)
(iii) A high ventilation rate provides more oxygen (1)
for the metabolism of lactic acid in the liver. (1)

(10 marks)

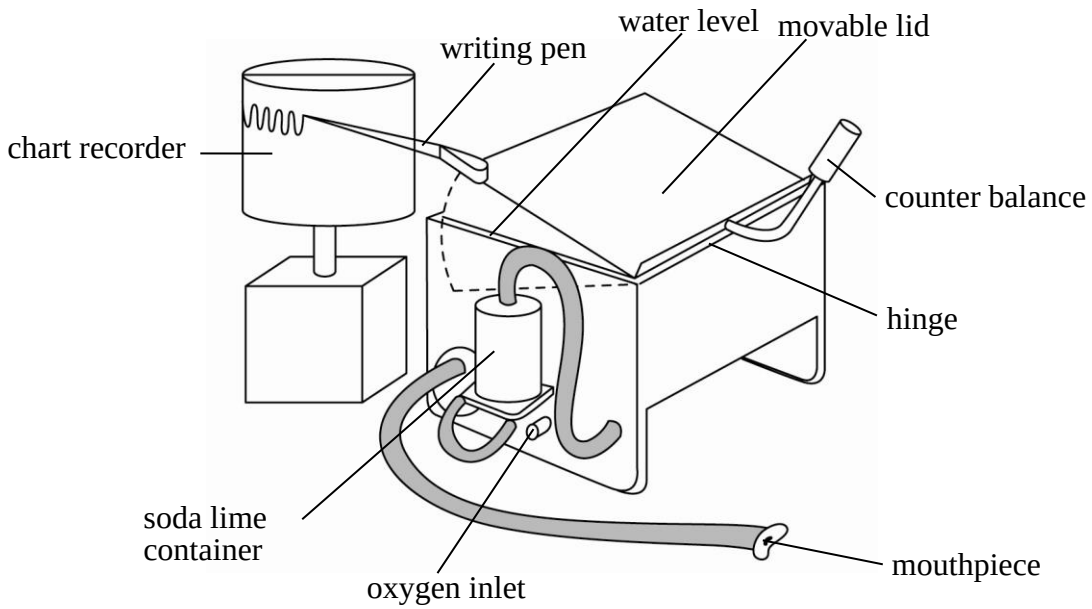
##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Analysing data	1
	Applying concepts	3
(c)(i)	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	1
(c)(iii)	Applying concepts	2

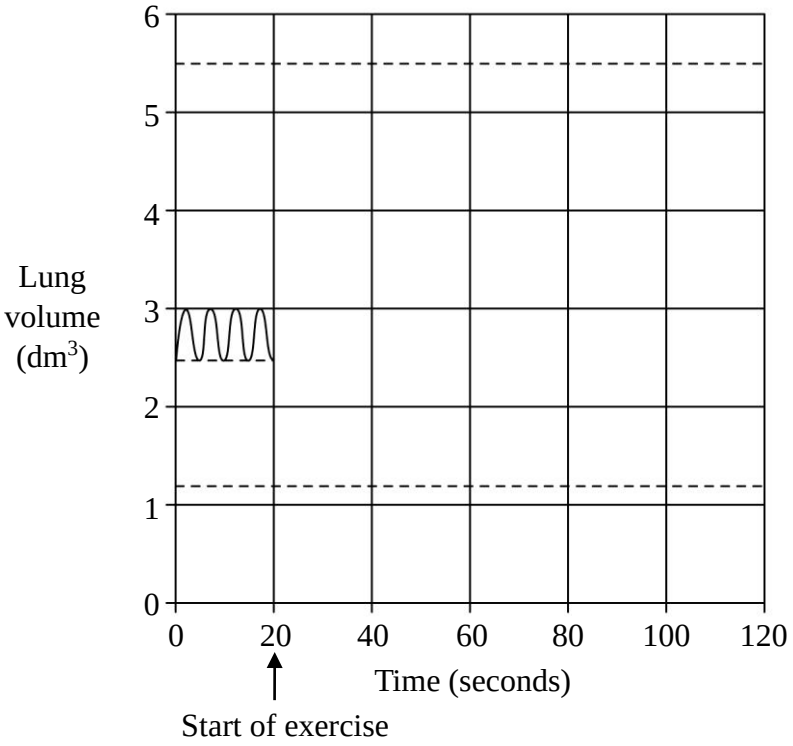
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|!|BCELA0340000008|!| ★★ **SI**

The apparatus shown below can be used to measure the tidal volume of humans. Tidal volume is the volume of air entering and leaving the lungs in each breath. A man was asked to breathe in and out repeatedly through the mouthpiece of the apparatus in order to measure his tidal volume.



The graph below shows the changes in the tidal volume of the man at rest.



(a) Suggest the use of soda lime in the apparatus. (2 marks)

(b) With reference to the graph,

(i) state the tidal volume when the man is at rest. (1 mark)

(ii) calculate his breathing rate at rest. (2 marks)

(c) Make a sketch on the graph to show how the tidal volume would change if the man started to do exercise at 20 seconds. (2 marks)

(d) Describe how the respiratory centre brings about the changes in (c). (3 marks)

##

(a) Soda lime absorbs carbon dioxide in the exhaled air. (1)
The air re-inhaled has no carbon dioxide. This can avoid the effect of carbon dioxide on the tidal volume. (1)

(b) (i) 0.5 dm^3 (1)

(ii) Breathing rate= (4 / 20) × 60 (1)
= 12 breaths per minute (1)

(c) Curve showing the rate of breathing increases (1)
and the depth of breathing increases (1)

(d) During exercise, the respiratory centre is stimulated send more nerve impulses through the sympathetic nerve to the intercostal muscles and diaphragm muscle. (1)
As a result, the intercostal muscles and diaphragm muscle contract
more frequently (1)
and more forcefully. (1)

(10 marks)

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##

	Skills assessed	Marks
(a)	Practical skills	2
(b)(i)	Analysing data	1
(b)(ii)	Mathematical skills	2
(c)	Applying concepts	2
(d)	Applying concepts	3

|!|BCELB0340000009|!| ★

The table below shows Aaron's condition at rest and during exercise.

	At rest	During exercise
Ventilation rate (cm^3 / min)	8 000	28 000
Volume of air inhaled per breath (cm^3)	500	1 000
Concentration of carbon dioxide in exhaled air (%)	4	4.5

(a) What are Aaron's breathing rates at rest and during exercise? (2 marks)

(b) State the importance of increasing the ventilation rate during exercise. (3 marks)

(c) Explain how the concentration of carbon dioxide in exhaled air can be kept more or less the same when Aaron is at rest and doing exercise. (5 marks)

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##

(a) Breathing rate at rest = 8000 ÷ 500 = 16 breaths per minute (1)
Breathing rate during exercise =28000 ÷ 1000 = 28 breaths per minute (1)

(b) More oxygen can be supplied to the skeletal muscle cells (1)
for the oxidation of more glucose to release more energy for muscle contraction. (1)
Carbon dioxide can be removed from the muscle cells more quickly. (1)

(c) More carbon dioxide is produced by skeletal muscle cells during exercise. (1)
Carbon dioxide in blood diffuses more rapidly into the air sacs. (1)
With an increased breathing rate, the amount of carbon dioxide diffusing from the blood into the air sacs per breath can be roughly the same, (1)
but more carbon dioxide can be breathed out per unit time. (1)
As a result, the carbon dioxide content of blood can be maintained at a normal level. (1)

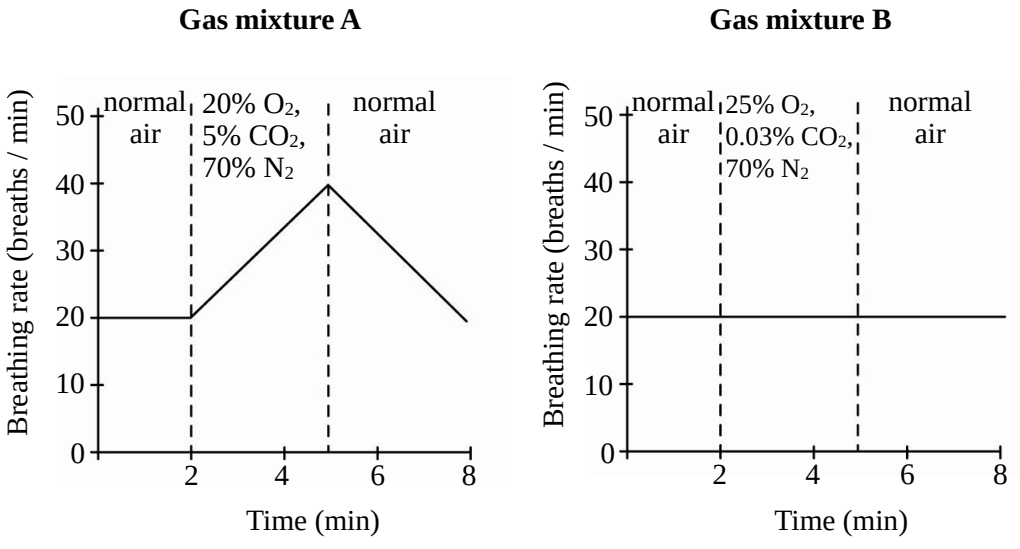
(10 marks)

##

	Skills assessed	Marks
(a)	Mathematical skills	2
(b)	Understanding basic concepts	3
(c)	Applying concepts	5

|!|BCELA0340000010|!| ★★

In an investigation, Sharon studied how the composition of inhaled air affects the breathing rate. She inhaled two gas mixtures, A and B, respectively. Her breathing rates were measured. The results are shown in the graphs below.



(a) With reference to the graphs, deduce oxygen or carbon dioxide has a greater effect on the breathing rate. (5 marks)

(b) In a traffic accident, a man stopped breathing. Based on your deduction in (a), suggest which of the following is a better method to help the man resumes breathing.

- perform mouth-to-mouth resuscitation
- force gas mixture B into his lungs using an appropriate instrument

(4 marks)

##

(a) Inhaling gas mixture B did not affect the breathing rate, (1)
implying that an increase in O_2 content (from 20% to 25%) did not affect the breathing rate. (1)
However, inhaling gas mixture A increased the breathing rate significantly, (1)
implying that an increase in CO_2 content (from 0.03% to 5%) led to an increase in the breathing rate. (1)
Therefore, carbon dioxide content has a greater effect on the breathing rate than oxygen content. (1)

(b) Performing mouth-to-mouth resuscitation is a better method. (1)
Exhaled air has a higher CO_2 concentration (~4%). (1)
The high concentration of carbon dioxide stimulates the respiratory centre of the patient to send nerve impulses to the intercostal muscles and diaphragm muscle. (1)
Gas mixture B contains a lower level of CO_2 (~0.03%), and is less effective in stimulating breathing. (1)

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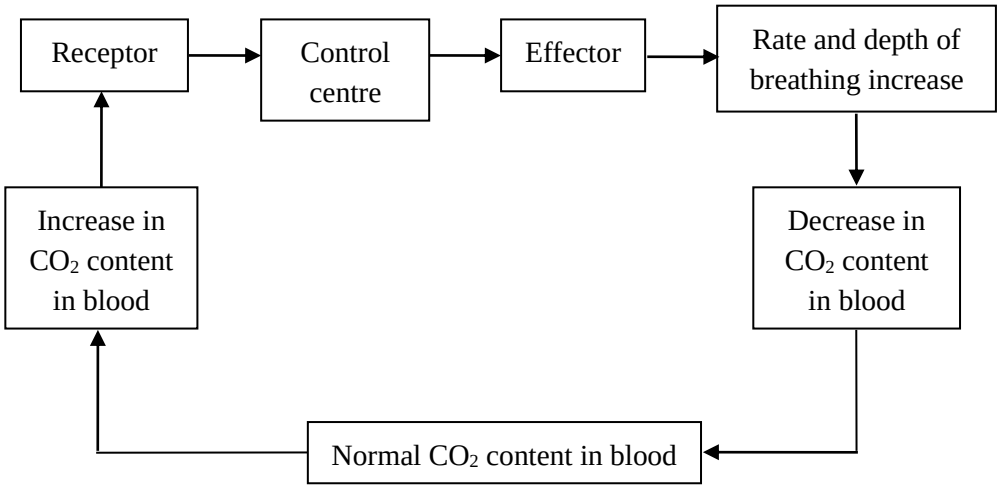
(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
	Applying concepts	3
(b)	Applying concepts	4

|!|BCELC0340000011|!|

The diagram below shows the regulation of carbon dioxide content in blood by controlling the rate and depth of breathing.



(a) Name the regulation mechanism as shown in the diagram. (1 mark)

(b) State the receptor and ***two*** of its locations in the body. (3 marks)

(c) State the control centre and the effector. How does the control centre lead to the response of the effector? (4 marks)

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##

(a) Negative feedback mechanism (1)

(b) The receptor is the chemoreceptor. (1)

Any ***two*** locations: (2)

- Carotid bodies
- Aortic bodies
- Respiratory centre in the medulla oblongata

(c) The control centre is the respiratory centre. (1)

The effectors are the intercostal muscles and diaphragm muscle. (1)

The respiratory centre sends more frequent nerve impulses to the intercostal muscles and diaphragm muscle, (1)

causing them to contract more frequently and more forcefully. (1)

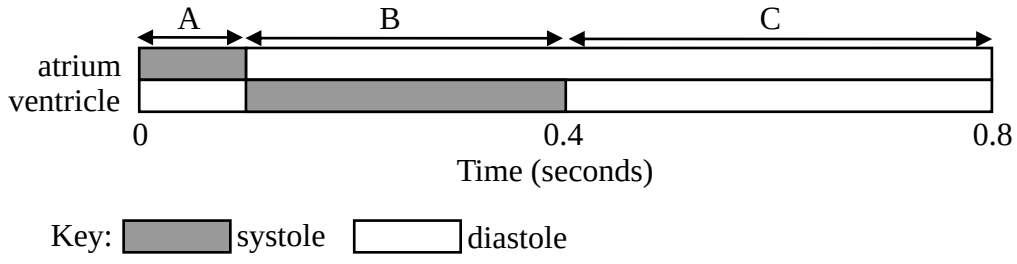
(8 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	3
(c)	Understanding basic concepts	4

|!|BCELC0340000012|!|

The diagram below shows a cardiac cycle which consists of three phases: A, B and C.



(a) Describe the states of the atria, ventricles and valves as well as the direction of blood flow in

(i) phase A (3 marks)

(ii) phase B (3 marks)

(iii) phase C (3 marks)

(b) When we are doing exercise, how will the duration of our cardiac cycle change? What is the significance of this change? (3 marks)

##

(a) (i) In phase A, the atria contract and the ventricles relax. (1)
The bicuspid valve and tricuspid valve are open, while the semilunar valves are closed. (1)
Blood flows from atria to ventricles. (1)

(ii) In phase B, the atria relax and the ventricles contract. (1)
The bicuspid valve and tricuspid valve are closed, while the semilunar valves are open. (1)
Blood flows from ventricles to pulmonary artery and aorta. (1)

(iii) In phase C, both the atria and the ventricles relax. (1)
The bicuspid valve and tricuspid valve are open, while the semilunar valves are closed. (1)

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Blood flows from the vena cava and pulmonary vein to atria; also from atria to ventricles. (1)

(b) During exercise, the heart rate increases, i.e. the duration of each cardiac cycle becomes shorter. (1)

An increased heart rate leads to faster blood circulation, hence oxygen and nutrients can be delivered to skeletal muscle cells more rapidly. (1)

Carbon dioxide can also be removed from the muscle cells more rapidly. (1)

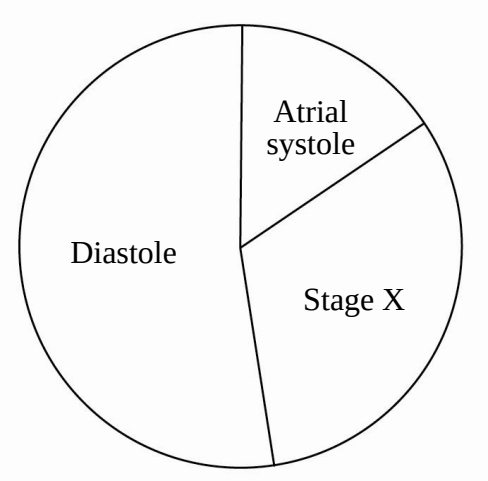
(12 marks)

##

	Skills assessed	Marks
(a)(i)	Understanding basic concepts	3
(a)(ii)	Understanding basic concepts	3
(a)(iii)	Understanding basic concepts	3
(b)	Applying concepts	3

|!|BCELC0340000013|!|

The cardiac cycle is the sequence of events that occurs in one heartbeat. It involves a sequential and coordinated contraction and relaxation of the four heart chambers. The following pie chart shows the relative lengths of three stages of a cardiac cycle.



(a) Calculate the length of one cardiac cycle if the heart beats in a rate of 75 times per minute. Show your working. (2 marks)

(b) What is the direction of blood flow within the heart during atrial systole? Explain your answer. (2 marks)

(c) Name stage X. State the condition of the bicuspid valve, tricuspid valve and the semilunar valves during stage X. (3 marks)

(d) How is the heart sound produced during diastole? (3 marks)

##

(a) Length of one cardiac cycle = 60 s ÷ 75 (1)
= 0.8 s (1)

(b) Blood flows from the atria into the ventricles. (1)
This is because when the two atria contract, the atrial pressure increases and becomes higher than the ventricular pressure. (1)

(c) Stage X is the ventricular systole. (1)
The bicuspid valve and tricuspid valve are closed. (1)
The semilunar valves are open. (1)

(d) During diastole, both the atria and the ventricles relax. (1)
When the pressure in the ventricles becomes lower than the pressure in the aorta and pulmonary artery, the semilunar valves close (1)
to prevent the backflow of blood from the two blood vessels into the ventricles. (1)
The closure of the semilunar valves produces the heart sound ‘dub’.

(10 marks)

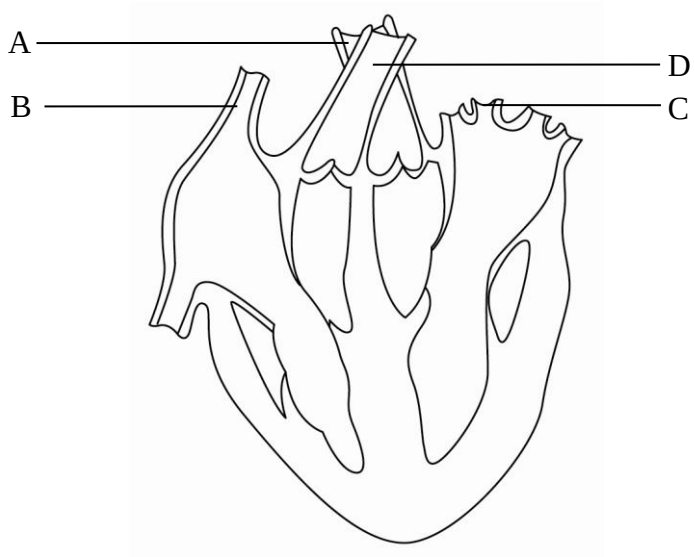
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	Skills assessed	Marks
(a)	Mathematical skills	2
(b)	Understanding basic concepts	2
(c)	Analysing data	1
	Understanding basic concepts	2
(d)	Applying concepts	3

|!|BCELC0340000014|!|

The diagram below shows the longitudinal section of a human heart at the shortest stage of the cardiac cycle.



(a) Suggest which stage of the cardiac cycle is shown in the diagram. Give ***two*** reasons to support your answer. (3 marks)

(b) Describe how this stage of cardiac cycle is initiated. (2 marks)

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(c) In which labelled blood vessel is the blood pressure the highest? Explain your answer. (3 marks)

##

(a) Atrial systole (1)
It is the shortest stage of the cardiac cycle. (1)
The bicuspid and tricuspid valves are open / The semilunar valves are closed. (1)

(b) The SA node / pacemaker generates electrical impulses which spread to both atria. (1)
This causes the atria to contract at the same time. (1)

(c) Blood vessel A (1)
Blood vessel A receives blood at high pressure pumped from the left ventricle when the left ventricle contracts. (1)
The walls of the left ventricle are thicker than the walls of the right ventricle, which produce a greater pumping force when the left ventricle contracts. (1)

(8 marks)

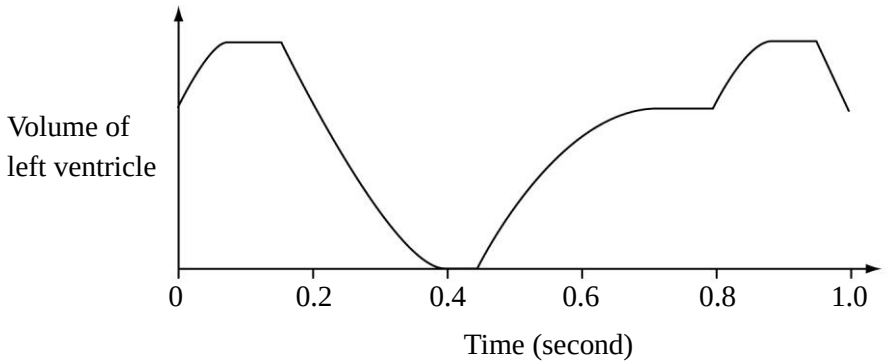
##

	Skills assessed	Marks
(a)	Analysing data	1
	Applying concepts	2
(b)	Understanding basic concepts	2
(c)	Applying concepts	3

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|!|BCELB0340000015|!| ★

The graph below shows the changes in the volume of the left ventricle of a man in one second.



(a) Calculate the heart rate of this man. Show your working. (2 marks)

(b) Suggest a reason to account for the decrease in the volume of the left ventricle between 0.2 second and 0.4 second. (1 mark)

(c) In view of your answer to (b), when the volume of the left ventricle is decreasing, describe the state of

(i) the bicuspid valve. (1 mark)

(ii) the semilunar valve in the aorta. (1 mark)

(d) Name the stage of the cardiac cycle

(i) during the first 0.05 second. (1 mark)

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(ii) between 0.45 and 0.8 second. (1 mark)

(e) How will the graph change when the man is doing vigorous exercise? Explain the significance of this change to skeletal muscle cells. (4 marks)

##

(a) 60 ÷ 0.8 (1)
= 75 beats per minute (1)
(b) The left ventricle is contracting. / Blood is leaving the left ventricle. (1)
(c) (i) The bicuspid valve is closed. (1)
(ii) The semilunar valve is open. (1)
(d) (i) Atrial systole (1)
(ii) Diastole (1)
(e) The duration of a cardiac cycle will become shorter / the heart rate will increase. (1)
The increase in heart rate allows faster supply of more blood / oxygen / nutrients (1)
to the skeletal muscle cells (1)
for cellular respiration to release more energy to support increased muscle contractions. (1)

(11 marks)

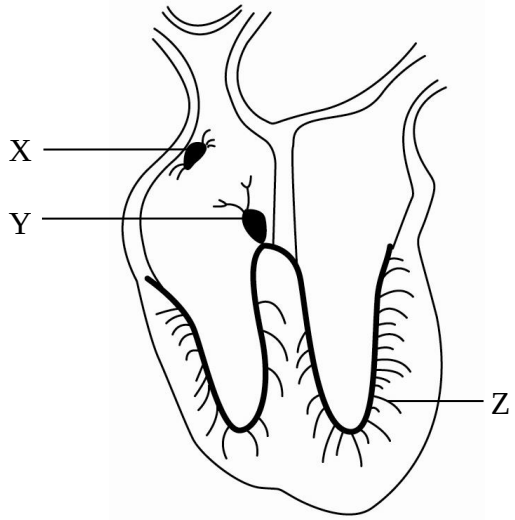
##

	Skills assessed	Marks
(a)	Mathematical skills	2
(b)	Applying concepts	1
(c)(i)	Applying concepts	1
(c)(ii)	Applying concepts	1
(d)(i)	Analysing data	1
(d)(ii)	Analysing data	1
(e)	Applying concepts	4

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|!|BCELB0340000016|!| ★

The diagram below shows the longitudinal section of a human heart and the structures involved in the initiation of heartbeat.



(a) Name the structures labelled X, Y and Z. (3 marks)

(b) Give ***one*** difference between the initiation of heartbeat and the initiation of skeletal muscle contraction. (2 marks)

(c) Describe the process of initiating a heartbeat, including the roles of structures X, Y and Z in the process. (5 marks)

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##

(a) X: sinoatrial node (SA node) (1)
Y: atrioventricular node (AV node) (1)
Z: Purkinje fibre (1)

(b) The heartbeat is initiated within the heart muscle while (1)
the contraction of skeletal muscles is initiated by nerve impulse from the brain or spinal cord. (1)

(c) Structure X generates electrical impulses which spread to both atria, (1)
causing both atria to contract. (1)
The electrical impulses then pass to structure Y which relays the electrical impulses along the septum (1)
to the base of the ventricles through the bundle of His. (1)
Structure Z conducts electrical impulses through the walls of ventricles, causing the ventricles to contract. (1)

(10 marks)

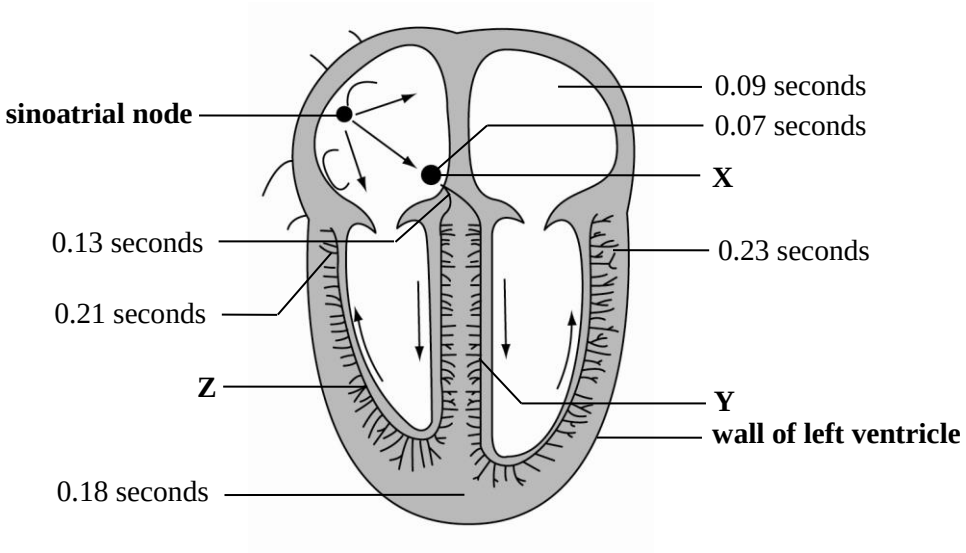
##

	Skills assessed	Marks
(a)	Analysing data	3
(b)	Applying concepts	2
(c)	Applying concepts	5

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|!|BCELB0340000017|!| ★

The diagram below shows the transmission of electrical impulses from the sinoatrial (SA) node to the rest of the heart through structures X, Y and Z. The times shown are relative to the time when an electrical impulse is initiated at the sinoatrial node.



(a) Name the structures labelled X, Y and Z. (3 marks)

(b) State and explain the stages of the cardiac cycle at

(i) 0.09 seconds. (2 marks)

(ii) 0.2 seconds. (2 marks)

(c) It takes a relatively longer time for the electrical impulses to be transmitted from structure X to the ventricles. What is the significance of this? (1 mark)

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(d) Explain why the electrical impulses should reach the base of the ventricles before going up the walls of the ventricles. (2 marks)

##

(a) X: atrioventricular (AV) node (1)
Y: bundle of His (1)
Z: Purkinje fibres (1)

(b) (i) Atrial systole (1)
The atria receive electrical impulse from the SA node and contract. (1)

(ii) Ventricular systole (1)
The electrical impulses have transmitted along the bundle of His to the ventricles, causing them to contract. (1)

(c) To allow sufficient time for the blood in the atria to flow into the ventricles. (1)

(d) This can cause the base of the ventricles to contract first (1)
so that the blood in the ventricles is squeezed up from the ventricles into the pulmonary artery and aorta. (1)

(10 marks)

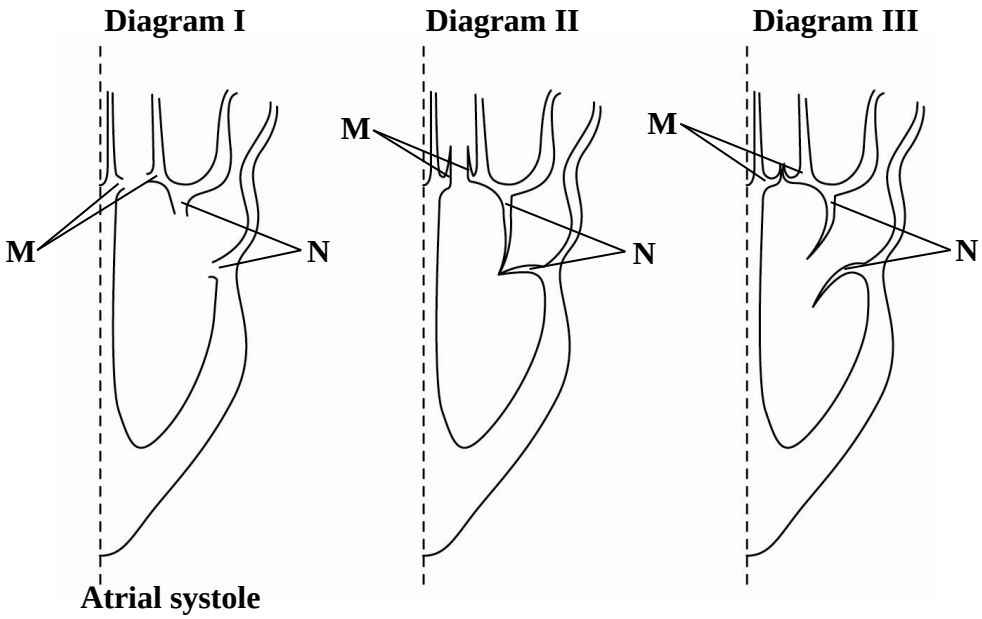
##

	Skills assessed	Marks
(a)	Analysing data	3
(b)(i)	Analysing data	1
	Applying concepts	1
(b)(ii)	Analysing data	1
	Applying concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	2

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|!|BCELB0340000018|!| ★

The diagrams below show the left heart in different stages of the cardiac cycle. M and N refer to the valves which are not drawn in Diagram I.



(a) Name valves M and N. (2 marks)

(b) Complete Diagram I by drawing valves M and N to show their states. (2 marks)

(c) With reference to the states of valves M and N, deduce the stage of the cardiac cycle represented by Diagram II. Explain your answer. (3 marks)

(d) Two heart sounds can be heard during a cardiac cycle. How are the two heart sounds produced? (4 marks)

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##

(a) M: semilunar valve (1)

N: bicuspid valve (1)

(b) M: closed (1)

N: open (1)

(c) Ventricular systole (1)

The closure of valve N / bicuspid valve suggests that the ventricular pressure is higher than the atrial pressure. (1)

The opening of valve M / semilunar valve suggests that the ventricular pressure is higher than the aortic pressure. (1)

Both occur when the ventricle is contracting.

(d) The first heart sound is produced by the closure of the bicuspid and tricuspid valves (1)

when the pressure in the ventricles becomes higher than that in the atria. (1)

The second heart sound is produced by the closure of the semilunar valves (1)

when the pressure in the ventricles becomes lower than that in the aorta and pulmonary artery. (1)

(11 marks)

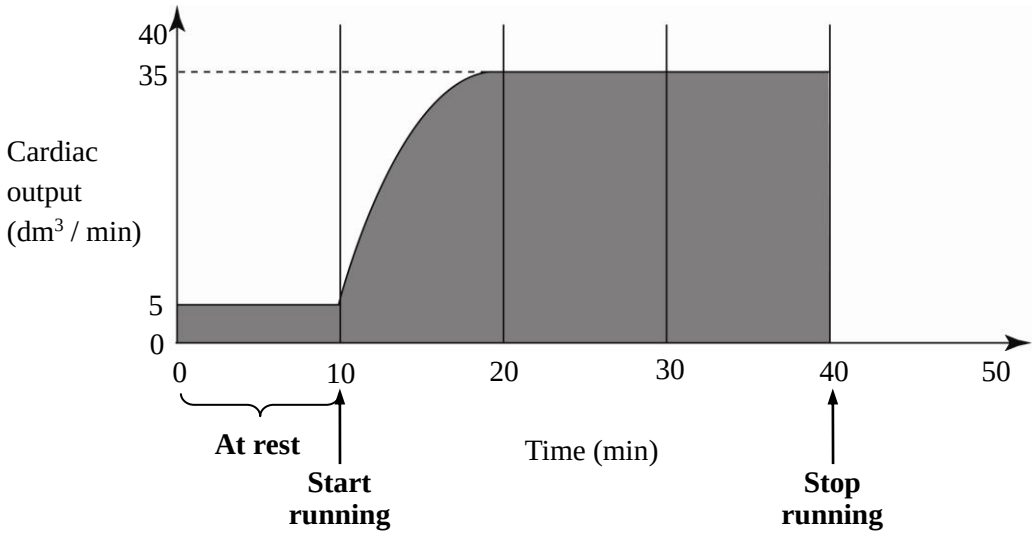
##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Applying concepts	2
(c)	Applying concepts	3
(d)	Understanding basic concepts	4

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|!|BCELC0340000019|!|

The graph below shows the changes in the cardiac output when a person was at rest for 10 minutes and then he ran on a treadmill for 30 minutes.



(a) Given that the heart rate of the person at rest was 80 beats per minute, calculate the stroke volume when the person was at rest. Show your working. (2 marks)

(b) How did the cardiac output change when the person was running on a treadmill? (1 mark)

(c) Describe how the change in (b) are controlled by the nervous system. (5 marks)

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(d) State the significance of the change in (b) to the person when he is running on a treadmill. (3 marks)

##

(a) Stroke volume = Cardiac output / Heart rate
= 5 dm^3 min^{-1} / 80 beats min^{-1} (1)
= 0.0625 dm^3 / beat (1)

(b) The cardiac output increased seven times. (1)

(c) During exercise, the carbon dioxide content in blood rises / blood pH lowers, (1)
this change is detected by the chemoreceptors in the carotid bodies and aortic bodies. (1)
These receptors send nerve impulses to the cardiovascular centre. (1)
This stimulates the cardiovascular centre to send more nerve impulses to the pacemaker via the sympathetic nerve. (1)
The cardiac muscles contract more frequently and forcefully, (1)
increasing the cardiac output.

(d) Blood circulation becomes faster. (1)
This helps transport oxygen and glucose more rapidly to the skeletal muscle cells for aerobic respiration. (1)
Carbon dioxide can also be transported away more rapidly from muscle cells to the lungs for faster removal. (1)

(11 marks)

##

	Skills assessed	Marks
(a)	Mathematical skills	2
(b)	Analysing data	1
(c)	Applying concepts	5
(d)	Applying concepts	3

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|!|BCELB0340000020|!| ★

The table below lists the heart rate and cardiac output of a person when he was doing exercise of different intensity.

Physical activity	**Heart rate (beats / minute)**	**Cardiac output (dm^3 / minute)**
At rest	70	5
Light	130	12
Moderate	150	16
Vigorous	180	20

(a) Calculate the stroke volume at rest. Show your working. (2 marks)

(b) Describe the effects of exercise of different intensity on the heart rate and cardiac output. (2 marks)

(c) Describe how the nervous system is involved in changing the heart rate and cardiac output when a person is exercising vigorously. (6 marks)

##

(a) Stroke volume = cardiac output ÷ heart rate
= 5 dm^3 min^{-1} ÷ 70 beats / min^{-1} (1)
= 0.071 dm^3 / beat (1)

(b) As the intensity of exercise increases, the heart rate increases (1)
and the cardiac output also increases. (1)

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(c) When a person is exercising vigorously, the rate of cellular respiration increases and carbon dioxide is produced at a faster rate. (1)
The carbon dioxide content of blood increases and blood pH decreases. (1)
This stimulates the cardiovascular centre in the medulla oblongata (1)
to send more nerve impulses through the sympathetic nerve to the SA node and cardiac muscle, (1)
causing the cardiac muscle to contract more frequently and more forcefully. (1)
As a result, the heart rate and stroke volume increases,
so does the cardiac output. (1)

(10 marks)

##

	Skills assessed	Marks
(a)	Mathematical skills	2
(b)	Analysing data	2
(c)	Applying concepts	6

|!|BCELB0340000021|!| ★ **Cross topic**

The table below shows the rate of blood flow to various body parts and the cardiac output at different levels of exercise.

Body part	Rate of blood flow (mL per minute)		
	At rest	Light exercise	Heavy exercise
Skeletal muscles	1200	4500	12500
Cardiac muscles	250	350	750
Brain	750	750	750
Skin	500	1500	1900
Cardiac output	5800	9500	17500

(a) Which body part has the greatest percentage increase in blood flow rate as the level of exercise increases? (1 mark)

(b) Explain the significance of the increase in the blood flow rate to the body part you have given in (a) during exercise. (4 marks)

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(c) Explain the change in the blood flow rate to the cardiac muscles when the intensity of exercise increases. (3 marks)

(d) Explain why an increase in blood flow to the skin is favourable to a person who is doing exercise. (3 marks)

##

(a) Skeletal muscles (1)

(b) Blood is supplied to the skeletal muscles at a higher rate. (1)
Oxygen and glucose can be supplied more rapidly to the skeletal muscle cells for carrying out aerobic respiration at a higher rate to release more energy for muscle contractions. (1)
Also, carbon dioxide can be removed more rapidly from the skeletal muscle cells. (1)

(c) As the intensity of exercise increases, the heart beats more frequently and forcefully. (1)
Increased blood flow to the cardiac muscles supplies more oxygen and glucose to the cardiac muscle cells. (1)
More energy is released by aerobic respiration to support the contraction of the cardiac muscles. (1)

(d) An increase in blood flow to the skin allows more heat to be lost (1)
through conduction, convection and radiation from the skin. (1)
This helps prevent overheating of the body during exercise. (1)

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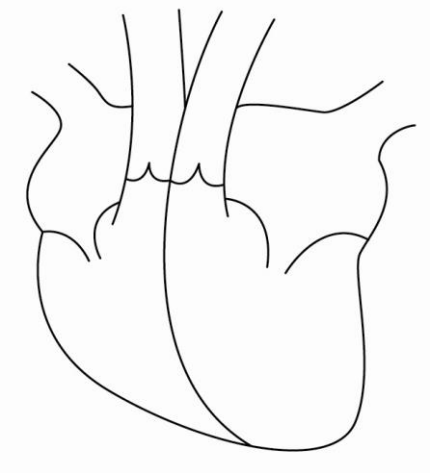
(11 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Applying concepts	4
(c)	Applying concepts	3
(d)	Applying concepts	3

|!|BCELB0340000022|!| ★ **STSE**

The diagram below shows a section of the heart.



(a) Draw arrows on the diagram to show the direction of blood flow during diastole. (1 mark)

(b) Explain why malfunctioning of the atrioventricular valves may cause exhaustion in the patient. (3 marks)

(c) Dysfunction of the sinoatrial (SA) node may cause irregular heart rhythm.

(i) What is the function of the sinoatrial (SA) node? (1 mark)

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(ii) Describe the series of events in one cardiac cycle starting from the sinoatrial node. (4 marks)

(iii) Suggest a treatment for SA node dysfunction. (1 mark)

##

(a) Arrows representing blood flow from veins through the atria to ventricles (1)

(b) If the atrioventricular valves do not close properly, some blood will flow from the ventricles to the atria during ventricular systole. (1)
Less blood flows into the aorta / pulmonary artery. (1)
Less blood, oxygen and nutrients can be supplied to body cells. (1)

(c) (i) It generates electrical impulses to initiate the contraction of cardiac muscle. (1)

(ii) The SA node generates electrical impulses which spread rapidly through the walls of both atria, causing both atria to contract at the same time. (1)
The electrical impulses then pass to the atrioventricular (AV) node which delays the electrical impulses for the ventricles to fill with blood. (1)
The AV node then relays the electrical impulses along the septum to the base of the ventricles. (1)
As the electrical impulses spread upwards through the walls of the ventricles, the ventricles contract from bottom up to squeeze blood upwards into the aorta and pulmonary arteries. (1)

(iii) Implantation of an artificial pacemaker (1)

(10 marks)

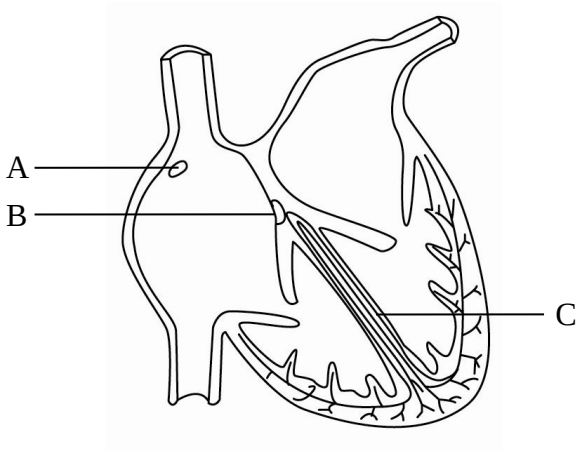
##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Applying concepts	3
(c)(i)	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	4
(c)(iii)	Understanding basic concepts	1

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|!|BCELC0340000023|!|

The diagram below shows a section through the heart.



(a) Name the structures labelled A, B and C. (3 marks)

(b) After structure A has generated electrical impulses, the ventricles contract after the atria have contracted. Explain why. (3 marks)

(c) What is the stage of cardiac cycle shown in the diagram? Explain your answer. (3 marks)

##

(a) A: sinoatrial node (SA node) (1)
B: atrioventricular node (AV node) (1)
C: bundle of His (1)

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(b) The electrical impulses from structure A spread to the atria directly, so they contract first. (1)

The electrical impulses from structure A has to pass through structures B and C to the base of the ventricles. (1)

The electrical impulses then spread upwards through the walls of the ventricles, causing the ventricles to contract from bottom up. (1)

Therefore, it takes a longer time for the ventricles to contract.

(c) Atrial systole / diastole (1)

Both the bicuspid valve and tricuspid valve are open, (1)

indicating that the atrial pressure is higher than the ventricular pressure. (1)

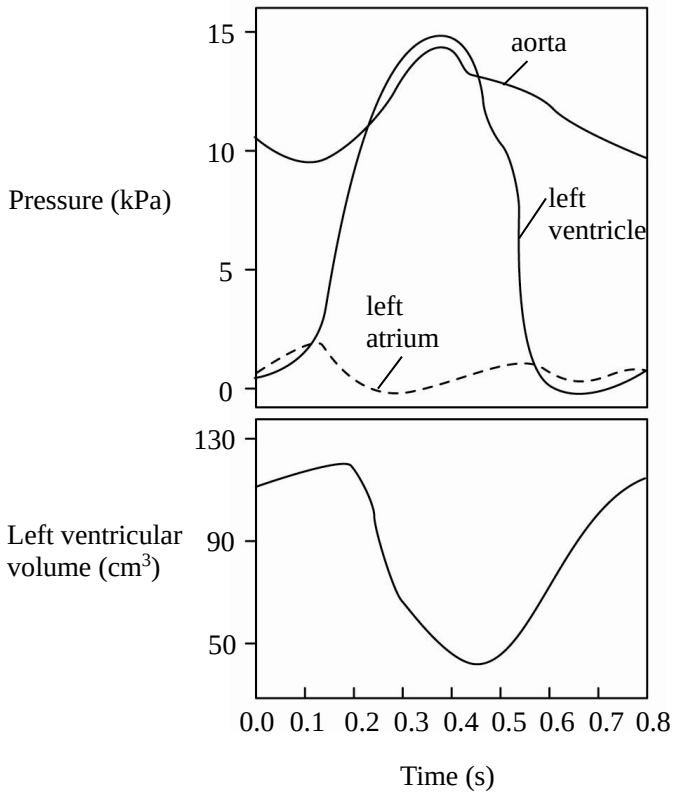
(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	3
(b)	Applying concepts	3
(c)	Analysing data	1
	Applying concepts	2

|!|BCELC0340000024|!|

The graph below shows the pressure changes in the left side of the human heart and the changes in the left ventricular volume during the cardiac cycle.



(a) With reference to graph, state and explain the changes of the following when the left ventricle contracts.

(i) Volume of the left ventricle (2 marks)

(ii) Pressure within the left ventricle (2 marks)

(b) How do the pressure changes in the left atrium, the left ventricle and the aorta affect the bicuspid valve and the semilunar valve at the base of the aorta? (4 marks)

(c) Describe how the two heart sounds are produced. (2 marks)

##

(a) (i) When the left ventricle contracts, the volume of the left ventricule drops from approximately 120 cm^3 to 40 cm^3. (1)
This is because the blood in the left ventricle is forced into the aorta when the left ventricle contracts. (1)

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(ii) When the left ventricle contracts, the pressure in the left ventricle rises to a peak of approximately 15 kPa. (1)

This is because the compression of the left ventricle decreases its volume and thus increases the pressure inside. (1)

(b) When the pressure in the left atrium is higher than that in the left ventricle, the bicuspid valve is open. (1)

When the pressure in the left ventricle is higher than that in the left atrium, the bicuspid valve is closed. (1)

When the pressure in the left ventricle is higher than that in the aorta, the semilunar valve is open. (1)

When the pressure in the aorta is higher than that in the left ventricle, the semilunar valve is closed. (1)

(c) The first heart sound is caused by the closure of the bicuspid valve during ventricular systole. (1)

The second heart sound is caused by the closure of the semilunar valves during diastole. (1)

(10 marks)

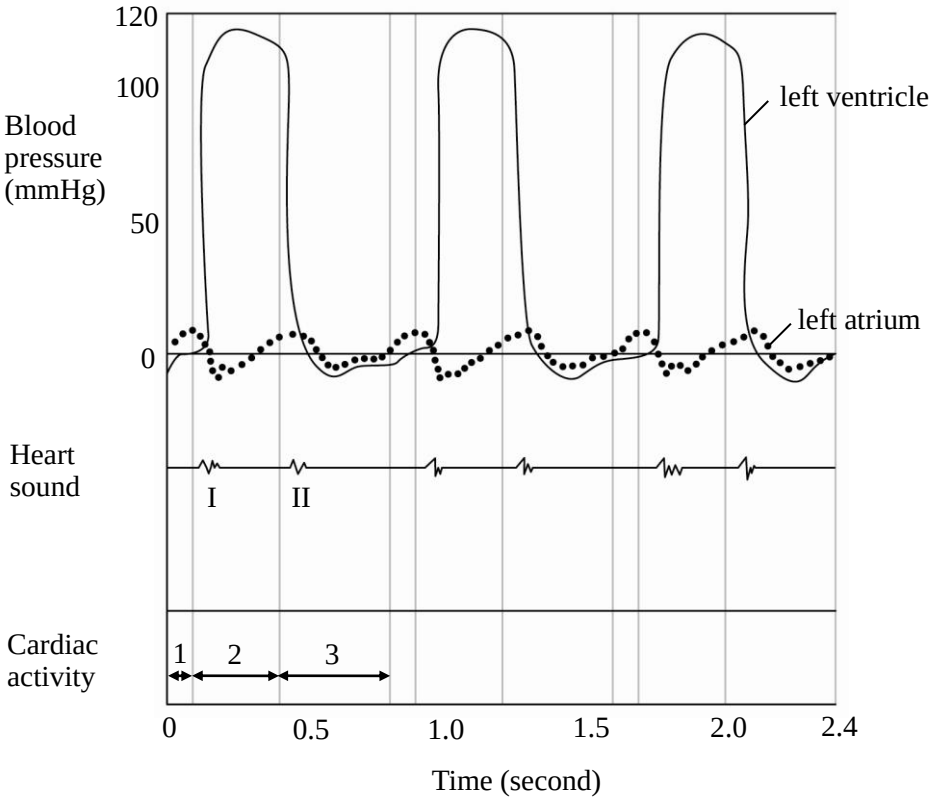
##

	Skills assessed	Marks
(a)(i)	Analysing data	1
	Applying concepts	1
(a)(ii)	Analysing data	1
	Applying concepts	1
(b)	Understanding basic concepts	4
(c)	Understanding basic concepts	2

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|!|BCELC0340000025|!|

The graph below shows the changes in the blood pressure in the left atrium and the left ventricle during three consecutive cardiac cycle of a man at rest.



(a) What does the difference between pressures in the left atrium and the left ventricle during cardiac activity 1 suggest about the direction of blood flow within the heart and the state of the atrioventricular valve? (2 marks)

(b) With reference to the graph, explain why heart sound I is produced during cardiac activity 2. (2 marks)

(c) Name cardiac activity 3. State one piece of evidence from the graph to support your answer. (2 marks)

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(d) What will be the change in the number of cardiac cycle over the same period of time if the person is doing vigorous exercise? (1 mark)

##

(a) Blood is flowing from the left atrium into the left ventricle. (1)
The atrioventricular valve is open. (1)

(b) During cardiac activity 2, the blood pressure in the left ventricle becomes higher than that in the left atrium. (1)
Hence, the bicuspid valve closes and this produces heart sound I. (1)

(c) Diastole (1)
The pressure of the left ventricle decreases sharply. (1)

(d) The number of cardiac cycle over the same period of time will increase. (1)

(7 marks)

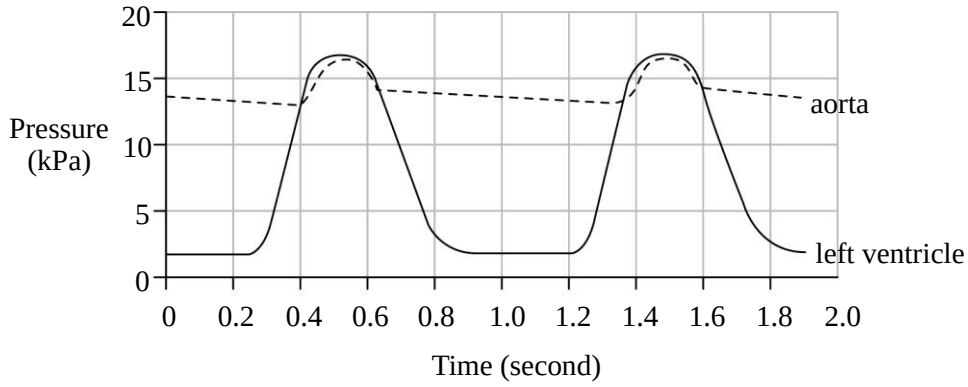
##

	Skills assessed	Marks
(a)	Analysing data	1
	Applying concepts	1
(b)	Analysing data	1
	Applying concepts	1
(c)	Analysing data	2
(d)	Applying concepts	1

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|!|BCELC0340000026|!|

The graph below shows the changes in the blood pressure in the aorta and the left ventricle during two consecutive cardiac cycles.



(a) State and explain the change in the blood pressure in the left ventricle from 0.3 to 0.5 seconds? (2 marks)

(b) At what time will the semilunar valves in the aorta open? Explain your answer. (3 marks)

(c) A heart sound can be heard at 1.6 seconds using a stethoscope. Explain how this sound is produced with reference to the graph. (3 marks)

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(d) How does the blood pressure in the right ventricle differ from that of the left ventricle? Suggest a reason for the difference. (2 marks)

##

(a) From 0.3 to 0.5 seconds, the blood pressure in the left ventricle increases (1)
as the left ventricle is contracting. (1)

(b) 0.4 seconds / 1.35 seconds (1)
At that time, the blood pressure in the left ventricle becomes higher than that of the aorta (1)
and the blood in the left ventricle begins to flow into the aorta. (1)

(c) At 1.6 seconds, the pressure in the left ventricle becomes just lower than that of the aorta. (1)
The semilunar valve at the base of the aorta closes (1)
to prevent the backflow of blood. (1)
This produces the heart sound.

(d) The blood pressure in the right ventricle would be lower. (1)
This is because the muscular wall of the right ventricle is thinner. (1)

(10 marks)

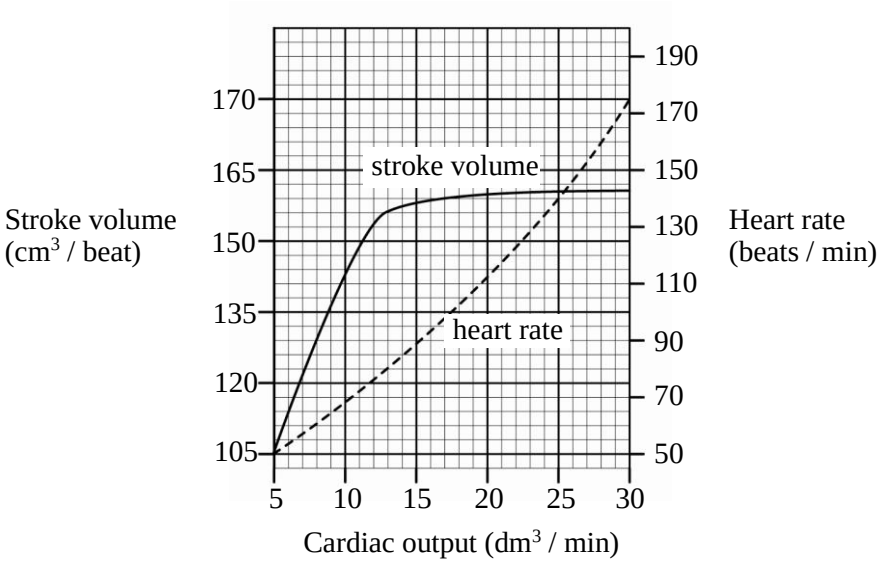
##

	Skills assessed	Marks
(a)	Analysing data	1
	Applying concepts	1
(b)	Analysing data	1
	Applying concepts	2
(c)	Analysing data	1
	Applying concepts	2
(d)	Understanding basic concepts	2

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|!|BCELB0340000027|!| ★ **SI**

Cardiac output can be defined as the volume of blood pumped out by each ventricle per unit time. During exercise, the heart rate and stroke volume are adjusted by the body, thereby varying the cardiac output. The graph below shows the relationship between the heart rate, the stroke volume and the cardiac output of a trained athlete.



(a) With reference to the above graph,

(i) what is the percentage change in his heart rate when the cardiac output is increased from 5 to 30 dm^3 / minute. Show your working. (2 marks)

(ii) describe and explain the changes in the heart rate and the stroke volume as the cardiac output increases from 5 to 30 dm^3 / minute. (6 marks)

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(b) How would the graph of an untrained person differ from the above graph of a trained athlete? (3 marks)

(c) State ***two*** control variables for an investigation about the effect of physical training on the maximum stroke volume. (2 marks)

##

(a) (i) Percentage change in the heart rate = (175 – 50) / 50 × 100% (1)
= 250% (1)

(ii) The heart rate increases greatly from 50 to 175 beats / min as the cardiac output increases from 5 to 30 dm^3 / minute. (1)
This is because the cardiac muscle contracts more frequently. (1)
The stroke volume increases from 105 to 159 cm^3 / beats as the cardiac output increases from 5 to 20 dm^3 / min. (1)
However, the stroke volume remains constant as the cardiac output increases from 20 to 30 dm^3 / min. (1)
This is because the cardiac muscle contracts more forcefully so that more blood can be pumped out in one heartbeat. (1)
The stroke volume stays constant when the maximum volume of blood that can be pumped out of the heart per beat is reached. (1)

(b) The heart rate of an untrained person would increase more rapidly than the trained athlete. (1)
The stroke volume of an untrained person would increase more slowly than the trained athlete and it would take longer time to reach the maximum. (1)
The maximum stroke volume of an untrained person would be lower than the trained athlete. (1)

(c) Any ***two*** of the following: (2)

- Length of time of investigation
- The initial fitness levels of the subjects before investigation
- The gender of the subjects
- Dietary intake of the subjects

(Or other reasonable answers)

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(13 marks)

##

	Skills assessed	Marks
(a)(i)	Mathematical skills	2
(a)(ii)	Analysing data	3
	Applying concepts	3
(b)	Applying concepts	3
(c)	Practical skills	2

|!|BCELB0340000028|!| ★

Figure I below shows the vertical section of the right side of the heart, which is at a certain stage of the cardiac cycle. Figure II shows the changes in the pressures in blood vessel 7 and heart chamber 3 within a period of time.

Figure I

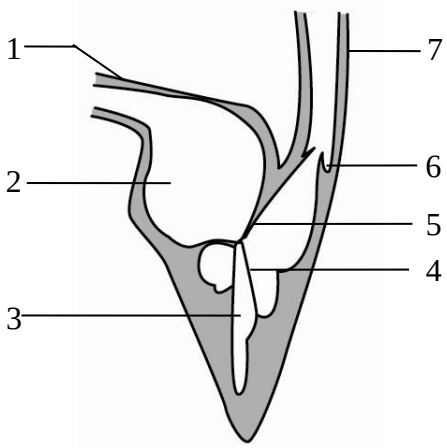
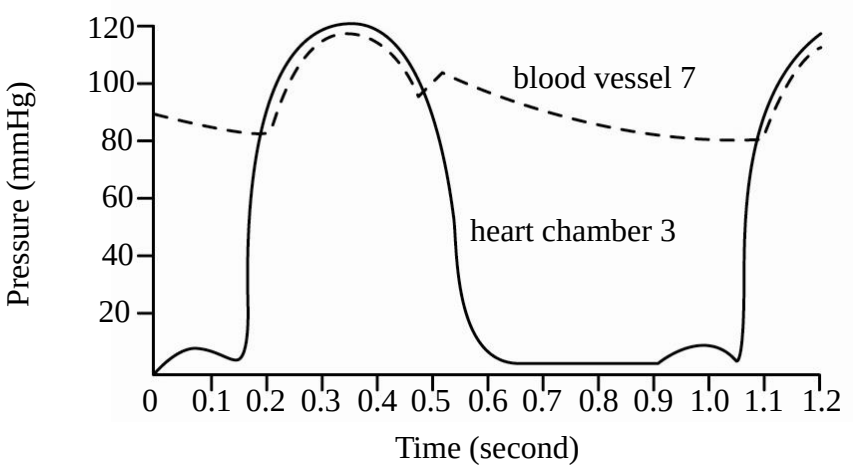


Figure II



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(a) Calculate the heart rate of this person. Show your working. (2 marks)

(b) During which time interval(s) in Figure II is the stage of the cardiac cycle shown in Figure I? Explain your answer. (3 marks)

(c) The cardiac output increases significantly during vigorous exercise. Explain the significance of this change. (4 marks)

##

(a) Duration of one heartbeat = 0.9 second (1)
Heart rate = 60 / 0.9 = 67 beats per minute (1)

(b) 0.2 to 0.5 seconds and 1.1 to 1.2 seconds (1)
Valve 6 is open in Figure I, (1)
implying that the pressure in chamber 3 is greater than that in vessel 7. (1)

(c) As a result of an increased cardiac output, the rate of blood circulation becomes faster. (1)
This helps to supply oxygen and glucose to skeletal muscle cells and cardiac muscle cells more rapidly so that these cells can carry out aerobic respiration at a faster rate to release more energy for muscle contractions. (1)
This also removes carbon dioxide and lactic acid from skeletal muscle cells and cardiac muscle cells more rapidly. (1)

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This also removes heat from skeletal muscles and cardiac muscle to prevent the muscles from overheating. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Mathematical skills	2
(b)	Analysing data	2
	Applying concepts	1
(c)	Understanding basic concepts	4